

BUILDING DESIGN
AI SOLUTION

CASE STUDY:

A High-rise Residential Condominium, Seoul

generate

Architect's Proposal

Type : Condominium

Phase : Feasibility Study

Date : March, 2021

Sales Price (per sq.ft.)

540 sq.ft. \$7,214

680 sq.ft. \$6,911

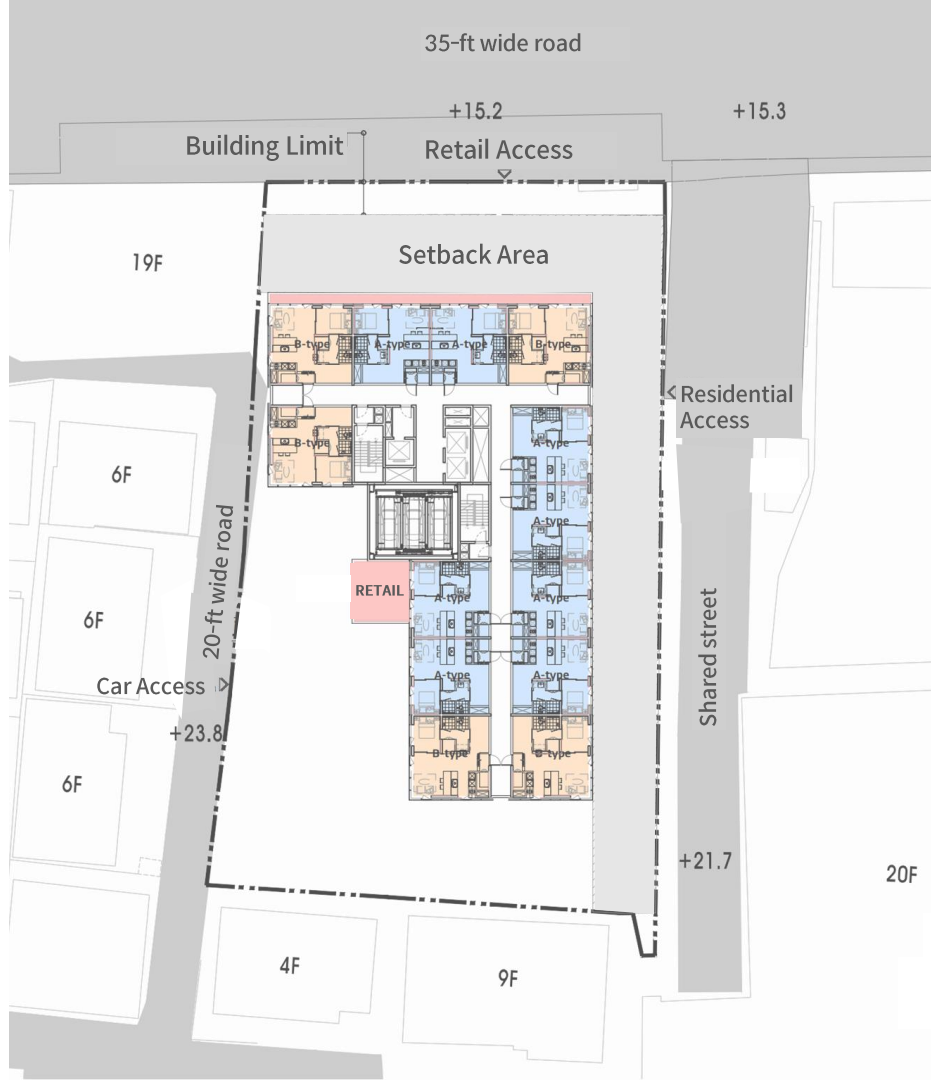
730 sq.ft. \$7,033

Construction Cost (per sq.ft.)

540 sq.ft. \$4,362

680 sq.ft. \$4,362

730 sq.ft. \$4,362



Sales Revenue

USD \$96.1 M

Building Efficiency

47.78%

FAR

6.59 / 6.60

Construction Cost

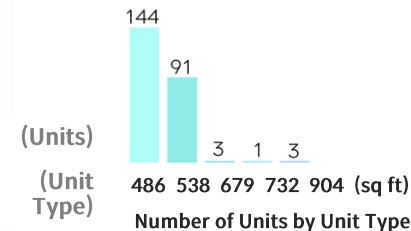
USD \$39.0 M

Number of Units

242

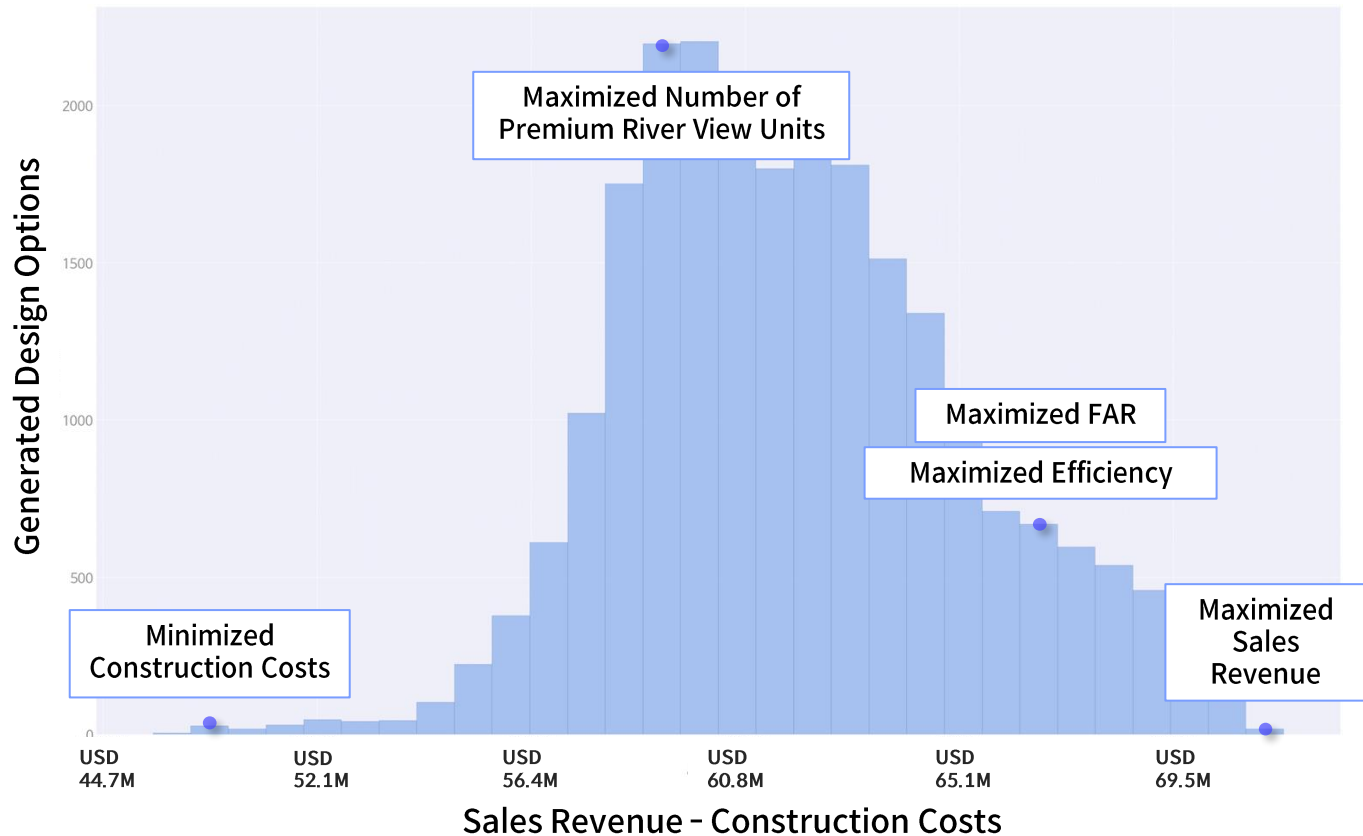
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B3 -20F



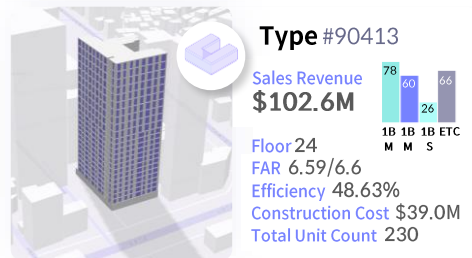
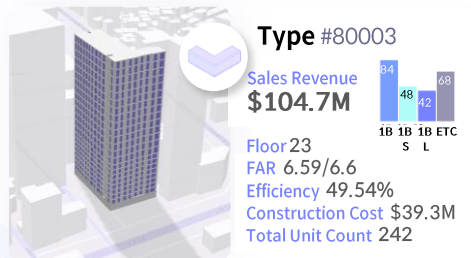
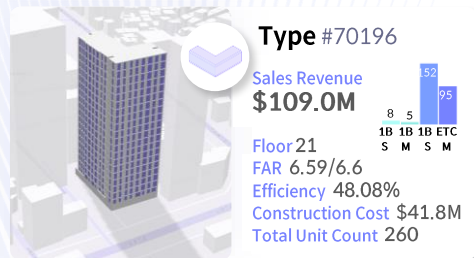
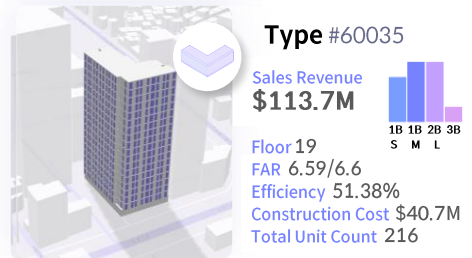
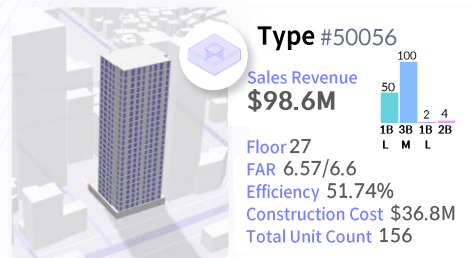
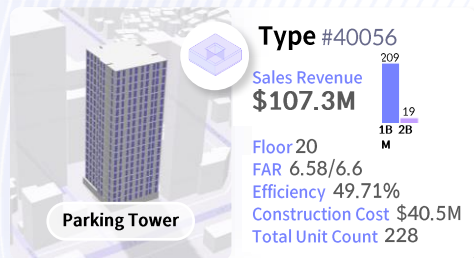
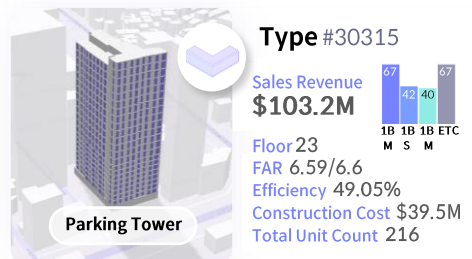
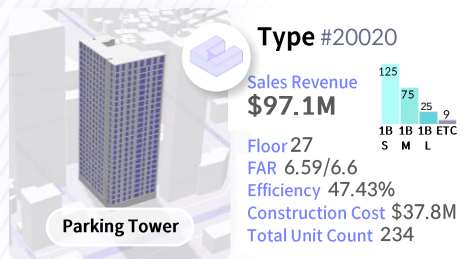
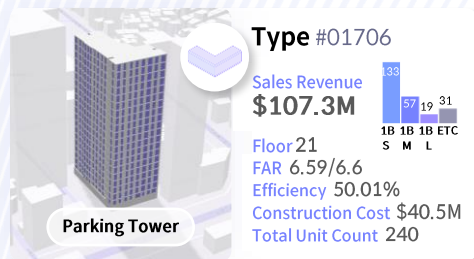
Zenerate™ AI Engine: 23,409 Design Options

- 9 Massing Types
- 10 Unit Types
(5 Loft Types)
- Sales Revenue:
USD 80.5M – 112.9M
- Construction:
USD 30.9M – 41.8M
- FAR:
5.76 - 6.60
- Efficiency:
45.17% - 54.17%
- Total Floors:
18 - 27



Massing Type Maximum Revenue

23,409 Solutions Based On 9 Massing Types



- UNIT TYPE MIX

1B/S	486 sq ft
1B/M	538 sq ft
1B/L	635 sq ft
2B	753 sq ft
3B	904 sq ft
1B/S	Loft 486 sq ft
1B/M	Loft 538 sq ft
1B/L	Loft 635 sq ft
2BL	Loft 753 sq ft
3BL	Loft 904 sq ft

Zenerate™ Analysis of Generated Design Costs Based On Parking Garage Types



Parking
Tower

11,140 Generated Designs

Parking Tower



Underground
Parking

12,269 Generated Designs

Underground Parking

vs

Sales Revenue	USD 97.7M		USD 102.3M	+ USD 4.6M ↑
Efficiency	50.54%	+0.46pp ↑	50.06%	
FAR	6.48		6.49	+ 0.01 ↑
Premium River View Units	104.8 Units	+19.2 Units ↑	85.6 Units	
Construction Costs	USD 37.5M		USD 38.9M	+ USD 1.4M ↑
Total Number of Units	209.3 Units		209.6 Units	+ 0.3 Units ↑
Total Floors	22.09 Floors	+1.2 Floors ↑	20.89 Floors	

Question:
Does Maximizing FAR Guarantee Maximized Revenue?

Answer: No, Maximizing FAR Does Not Guarantee Maximized Revenue

FAR average 6.53 for Top 5% (\$108.8M+) of Sales Revenue
Design Options Exceeding 6.58% FAR Ranged from Sales Revenue of \$94.8M to \$112.9M (\$18.1M Range)



The increase in FAR is somewhat proportional to sales revenue, but a higher FAR does not guarantee higher sales revenue.



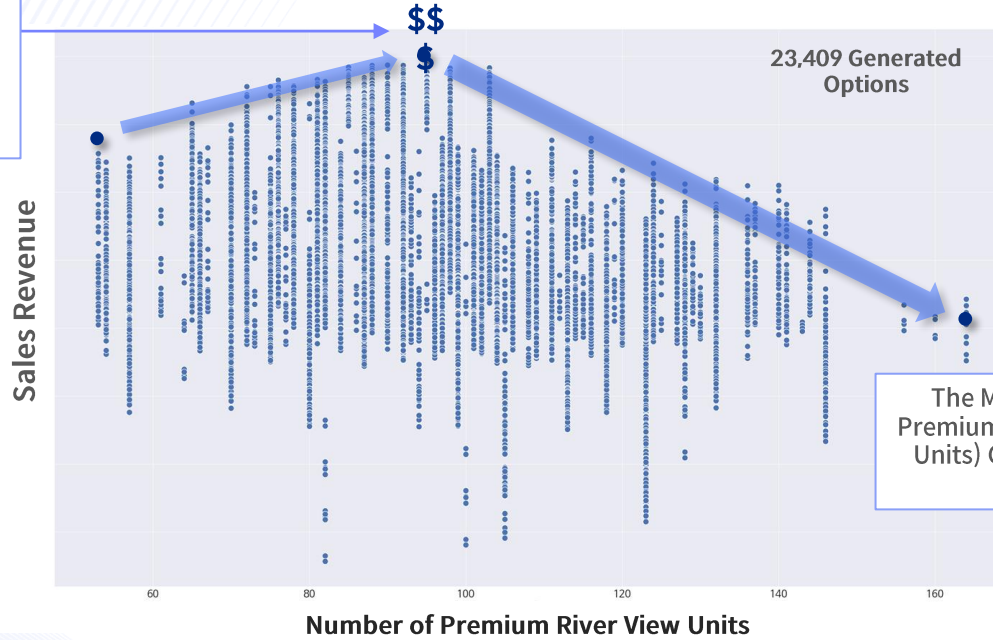
Question:
Does Maximizing Number of Premium River View Units
Maximize Sales Revenue?

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Answer: No, Maximizing Premium River View Units Does Not Maximize Sales Revenue

Although increasing the number of premium river views leads to higher sales revenue, sales revenue tends to decrease after a certain level. In this scenario, sales revenue drops for design options with more than 100 premium river view units.

Designs with Fewer Than 100 Premium River View Units Will Maximize Sales Revenue



The Maximized Number of Premium River View Units (160+ Units) Generates Lower Sales Revenue

Analysis of Precise Unit Views and Respective Sales Revenue

1. Utilizing drones, we measured heights at which river views were possible for each unit location (river view, partial view, no view)

2. Sales calculations based on river view unit types

Based on river view sales data (17 Residential Building)

Zenerate™ AI-integrated Pricing Data

Sales Data
(518 Sales)

Public Sales Data
(5,156 Sales)

Real-time Data
(275 Sales)

We used our AI engine to learn prices from actual transaction data for this type of building and location.

→ Consolidated each data type to improve accuracy

A1 – River View starts at 196.85 feet

Premium River View (Front)



B2 – River View starts at 196.85 feet

Partial River View



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Question:

Does Maximizing Efficiency Guarantee Maximized Revenue?

Answer: No, Maximizing Efficiency Does Not Guarantee Maximized Revenue

Efficiency averaged **50.67%** for the top 5% (\$108.8M+) of the sales volume.
Efficiency was more than 52% of the total sales rate for more than 15% of the total.

#Case60035



Sales Revenue \$102.9 M
Efficiency 51.38%

#Case1388



Sales Revenue \$103.7 M
Efficiency 53.73%



Question:

Do Loft Units Maximize Sales Revenue in this Site?

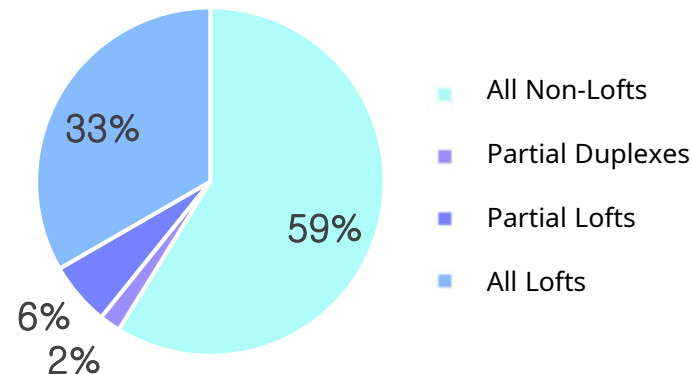
Answer: Yes, All Lofts is the best option in this site

<Average Price of Unit Types From 23,409 Zenerate™ Options>

Unit Mix	Loft	Loft Non-Loft	Non-Loft Loft	Non-Loft
	Lofts (1,425)	Lofts/ Non-Lofts (7,387)	Non-Lofts/ Lofts (6,896)	Non-Lofts (3,786)
Average Sales Revenue	\$109.1 M	\$103.5 M	\$98.3 M	\$96.0 M
Average Construction	\$40.7M	\$39.2M	\$37.8M	\$37.0M

< Current Status of Apartments with Lofts/Duplex >

Distribution of unit mix currently on the market (Surveyed area: Total number of sales in Seoul, 2021.06.02)



Twenty-one of the 51 cases currently on sale are lofts/duplexes, 17 of which are lofts with attic layouts. In areas where the height of the loft attic is possible, the trend to maximize sales is to build loft units.

An isometric illustration of a cityscape with various skyscrapers and buildings in shades of blue and purple. A winding road or path leads through the city. The background features a large, light blue circular shape and a series of concentric, wavy lines on the left side.

Question:

The highest revenue generating option by Our Zenerate™ AI Engine

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Zenerate™ Maximized Revenue

Compared to Architect's Plan
Sales Revenue Increased

+\$17M Boost

<How Zenerate maximized revenue>

1. Minimized corridor areas of corners
2. Increased floor area per floor
→ decreased number of floors
→ decreased size of common areas
3. Changed parking tower to Underground parking
→ increased efficiency rate
4. Adopting massing type and layout
that increase number of premium river views

Analysis of Sales Transactions

Unit Type	Sale	Rent
431 ~ 646 sq ft	17%	27%
646 ~ 915 sq ft	9%	5%
915 ~ 1076 sq ft	15%	4%



Type #60035

Sales Revenue \$113.1M

Floor 19

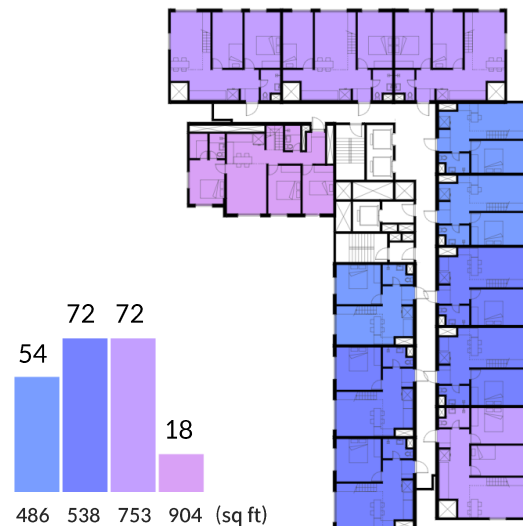
FAR 6.59/6.6

Efficiency 51.38%

Construction Cost \$41.2M

Total Unit Count 216

<2ND Floor Plan>



Zenerate™ Maximized Revenue

Compared to Architect's Plan
Sales Revenue Increased

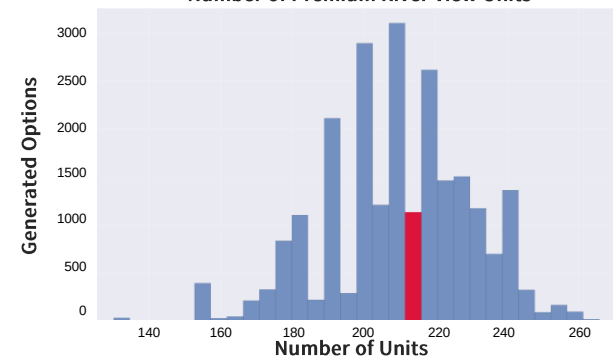
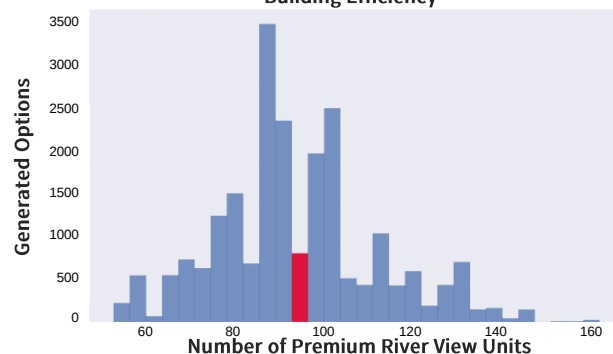
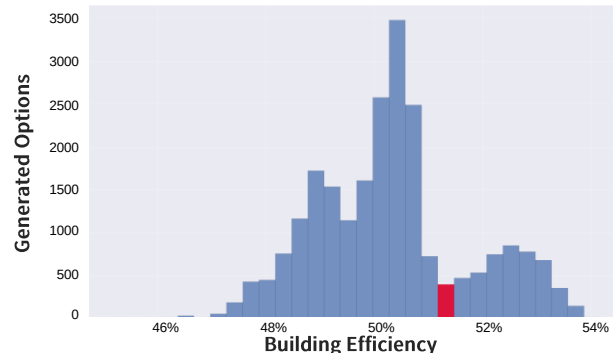
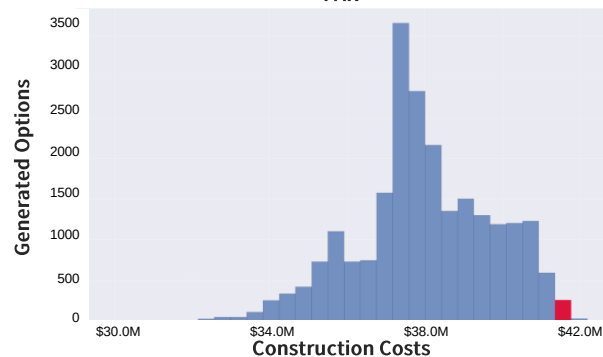
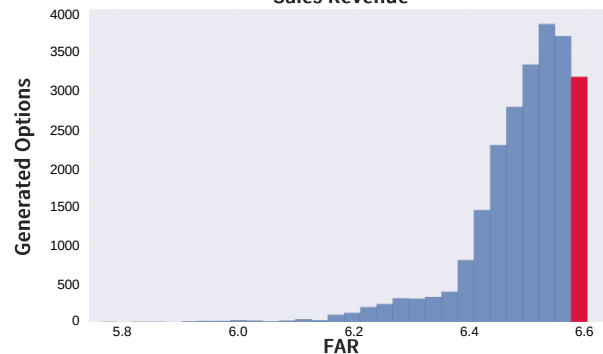
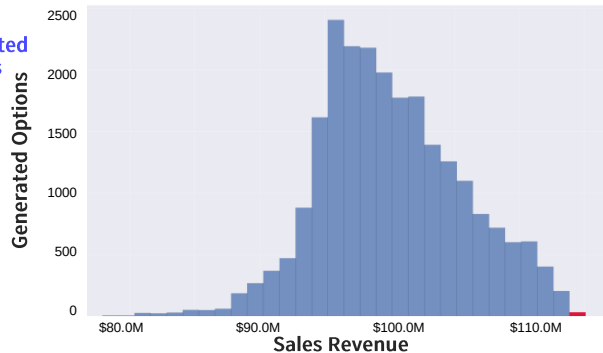
+\$17M Boost

19	Loft Units
18	Loft Units
17	Loft Units
16	Loft Units
15	Loft Units
14	Loft Units
13	Loft Units
12	Loft Units
11	Loft Units
10	Loft Units
9	Loft Units
8	Loft Units
7	Loft Units
6	Loft Units
5	Loft Units
4	Loft Units
3	Loft Units
2	Loft Units
1	Retail

Height of
Loft Attic:
14.1 ft

Section Diagram

23,409
Generated
Designs



Zenerate™ Maximized Revenue

Compared to Architect's Plan
Sales Revenue Increased

+\$17M Boost

	Existing Plan	Zenerate™ Plan	Variance
Sales Revenue	\$96.1 M	\$113.1 M	▲ +\$17.0 M
Building Efficiency	47.78%	51.38%	▲ +3.6pp
FAR	6.5955	6.5995	▲ +0.004
Construction Costs	\$39.0 M	\$41.2 M	▲ +\$2.2 M
Number of Units	242 Units	216 Units	▼ -26 Units
Total Floors	20 Floors	19 Floors	▼ -1 Floor

The background of the slide is a solid blue color. It features several large, overlapping circles in different shades of blue. Additionally, there are fine, concentric wavy lines that create a textured, ripple-like effect across the entire background.

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